Alberta Public Health Disease Management Guidelines

Hepatitis B – Acute and Chronic

This publication is available online at https://open.alberta.ca/publications/hepatitis-b-acute-and-chronic

For further information on the use of this guideline contact:

Health.CD@gov.ab.ca

Health and Wellness Promotion Branch

Public and Rural Health Division

Alberta Health

Hepatitis B – Acute and Chronic | Alberta Health, Government of Alberta

Table of Contents

Revisions

Revision Date	Document Section	Description of Revision
June 2024	Appendix 1	• Added new Appendix 1: Hepatitis B Virus Infection – High Endemic Geographic Areas

Case Definition

Acute Case	
Confirmed	• Laboratory confirmation of infection with clinical illness[(A)] or probable exposure within the last 6 months: - Immunoglobulin M antibody to Hepatitis B core antigen (anti-HBc IgM) positive **AND** one of the following: ▪ Hepatitis B surface antigen (HBsAg) positive[(B)] ***or*** ▪ HBV DNA positive[(B)] **OR** • Clearance of HBsAg within a 6 month period in a person who was documented HBsAg positive with a history of clinical illness[(A)] or probable exposure.
Probable	Acute clinical illness[(A)] in a person who is epidemiologically linked to a confirmed case (acute or chronic) and test results are unknown /unavailable

Chronic Case	
Confirmed	Laboratory confirmation of infection with or without clinical illness[(A)]: • Detection of Hepatitis B surface antigen (HBsAg)[(B)] ***or*** HBV DNA[(B)] ***or*** HBeAg for more than 6 months; **OR** • Immunoglobulin M antibody to Hepatitis B core antigen (anti-HBc IgM) negative **AND** at least one of the following: - HBsAg positive[(B)] ***or*** - HBV DNA positive[(B)] ***or*** - HBeAg positive **OR** • Total antibody to Hepatitis B core antigen (anti-HBc total) positive **and** HBV DNA positive; **and** - HBsAg negative and Antibody to Hepatitis B Surface Antigen (anti-HBs) negative
Probable	Laboratory confirmation of infection: • Single HBsAg positive in the context of: - history of compatible clinical illness more than 6 months ago; ***or*** - self-reported history of Hepatitis B testing and/or diagnosis more than 6 months ago; ***or*** - born and/or lived in Hepatitis B endemic country (prevalence ≥8%) more than 6 months ago.

[(A)] Clinical illness: a discrete onset of symptoms (e.g., fever, headache, malaise, anorexia, nausea, vomiting, abdominal pain, dark urine) and either jaundice or elevated serum aminotransferase level.

[(B)] Dried Blood Spot (DBS) testing is being used in some parts of Alberta and can be a valuable tool to assist in diagnosing Hepatitis B Virus (HBV), especially in remote communities and in hard-to-reach populations. DBS testing performed at an accredited laboratory is considered an appropriate clinical specimen for the confirmation of HBV infection.

Reporting Requirements

Physicians/Health Practitioners and Others

- Physicians, health practitioners, and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases in the prescribed form by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

- All laboratories shall report all positive laboratory results by mail, fax, or electronic transfer within 48 hours (two business days) to:
  - the MOH (or designate) of the zone and
  - the Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within four weeks of notification and the final NDR (amendment) within ten weeks of notification.
- For out-of-province and out-of-country cases/contacts, the following information should be forwarded to the CMOH (or designate) by phone, fax, or electronic transfer within 48 hours (two business days) including:
  - name,
  - date of birth,
  - out-of-province health care number,
  - out-of-province address and phone number,
  - positive laboratory report (if applicable), and
  - other relevant clinical/epidemiological information.

Additional Reporting Requirements

Canadian Blood Services (CBS)

- All confirmed hepatitis B cases must be reported by the MOH (or designate) to CBS within two business days if:
  - the person has a history of donating blood/blood products ***or***
  - the person received blood/blood products in Canada when blood transfusion is the only risk factor identified.
- A copy of the positive test result must accompany the Transmissible Disease Notification (TDN) form, and all information should be sent for Lookback/Traceback to the TDN Department by:
  - Fax 1-844-836-6843 **OR**
  - Scan and email to: TDnotifications@blood.ca
- To speak to a TDN Specialist regarding any questions or concerns please call **(506) 648-5076** or **1 (888) 992-5663** ext. 5076.

Clinical Assessment and Epidemiology

Etiology

The hepatitis B virus (HBV) is a DNA virus, in the hepadnavirus family. It consists of an inner nucleocapsid core (the hepatitis B core antigen [HBcAg]) which is surrounded by an outer lipoprotein coat containing the hepatitis B surface antigen (HBsAg).[1,2] The virus primarily infects liver cells (hepatocytes are the target cells) and causes both acute and chronic HBV infections.[3]

Clinical Presentation

Individuals infected with acute HBV infection may or may not have symptoms. Less than 10% of children, and 30–50% of adult, acute cases will have icteric disease. In persons with clinical illness, onset is usually subtle with initial symptoms of vague abdominal discomfort, loss of appetite, nausea, vomiting, sometimes arthralgia, and rash. Jaundice may develop after this initial phase of symptoms. Illness in acute cases may last up to three months. The case-fatality rate is about 1–2% and is higher in those over 40 years of age.[1,4]

Most adults with acute HBV infection recover completely and produce neutralizing antibodies, resulting in immunity from future infection. Following acute HBV infection, the risk of developing chronic infection varies inversely with age. As many as 90% of infants infected at birth or during the first year of life, as well as 25–50% of children infected between one and five years of age, will develop chronic infection. The risk of developing chronic infection in persons infected as older children and adults is approximately 5–10%. In addition, individuals with underlying chronic conditions or who are immunocompromised, have an increased risk of developing chronic HBV. Chronic HBV cases have an increased risk of developing hepatocellular carcinoma and cirrhosis. [1,3,5]

Diagnosis

Different serologic markers are used to determine if a person has an acute or chronic HBV infection. A dried blood spot (DBS) test offers an alternative, simplified specimen collection method that allows for Hepatitis B diagnosis in remote areas and/or for use in populations where access to serological testing is limited. DBS testing uses blood from a fingerstick procedure collected on filter paper and allowed to dry. DBS testing, performed at the National Microbiology Laboratory (NML)), detects HBsAg and HBV DNA and provides confirmation of HBV infection.[6–8]

For more information, refer to the following:

- Table 1: Hepatitis B Serological Markers
- Figure 1: Typical Course of Hepatitis B: Acute with Recovery and Chronic
- Table 2: Interpretation of Serologic Test Results for HBV

Table 1: Hepatitis B Serological Markers[9,10]

Positive Marker	Interpretation[1,3]
HBsAg Hepatitis B surface antigen	• This is the first detectable and primary marker of HBV infection • Can be detected in serum from 1-2 weeks to 11–12 weeks after exposure, or indefinitely in chronic infections • Transiently seen shortly after immunization (i.e., within 30 days following a dose of hepatitis B vaccine)
Anti-HBs Antibody to hepatitis B surface antigen	• Indicates immunity either from infection or vaccine
Anti-HBc IgM Immunoglobulin M (IgM) antibody to hepatitis B core antigen	• Is present in high titres in acute cases and usually disappears within six months • May sometimes be observed in exacerbations of chronic infection
Total Anti-HBc Antibody to hepatitis B core total	• Appears at the onset of symptoms in acute infection and persists for life • Indicates past or current infection • Detects both anti-HBc IgM and anti-HBc IgG • Not present after immunization
HBeAg Hepatitis B envelope antigen	• Sometimes present in infected individuals • Associated with high levels of HBV replication and potential for transmission (infectivity) • Can be present in both acute and chronic infections
HBV DNA	• Measures level of circulating DNA and is a marker of infectivity • Usually used to determine need for treatment and response to treatment • Not routinely performed during public health follow up

Figure 1: Typical Course of Hepatitis B: Acute with Recovery (L) and Chronic (R) [11]

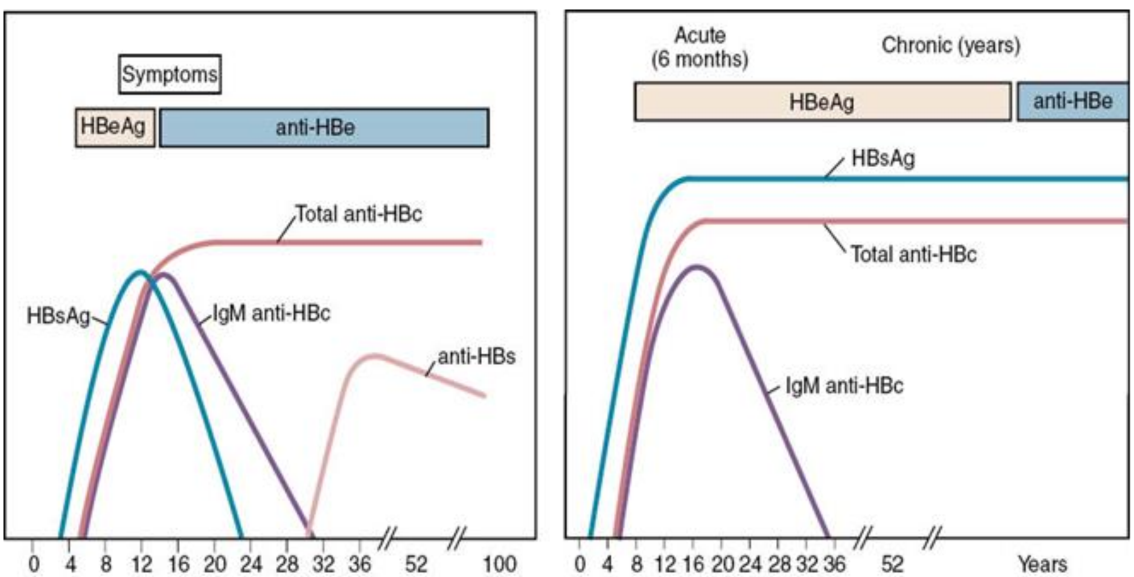

Table 2: Interpretation of Serologic Test Results for HBV[9,10,12]

HBsAg	Anti-HBc IgM	Total Anti-HBc	Anti-HBs	HBV DNA	Interpretation
+	-	-	-	n/a*	Early HBV infection before anti-HBc response, or transiently seen in early response to immunization (i.e., within 30 days following a dose of hepatitis B vaccine)
+	+	+	-	n/a	• Early HBV infection • Since anti-HBc IgM is positive, the onset is generally within six months • IgG antibody usually appears shortly after IgM; therefore, both are usually positive when IgM is positive (**acutely infected**)
-	+	+	- or +	n/a	• Recent acute HBV infection (within four to six months) with resolution, i.e., HBsAg has already disappeared • Anti-HBs usually appears within a few weeks or months of HBsAg disappearance
+	-	+	-	n/a	• HBV infection onset at least six months earlier because Anti-HBc IgM has disappeared • Probable chronic HBV infection (**chronically infected**)
-	-	-	+	n/a	• Response to hepatitis B vaccine • No evidence of infection
- or +	-	+	-	+	Chronic HBV infection
-	-	+	+	n/a	Past HBV infection, recovered (immune due to natural infection)
-	-	+	-	-	Four possible interpretations may be: ** • recovering from acute infection • distantly immune from past infection and the test is not sensitive enough to detect a very low level of Anti-HBs • a false positive Anti-HBc (remains susceptible to Hepatitis B) • chronically infected and have undetectable level of HBsAg

* *n/a = not routinely performed as part of public health follow-up*
****** *Consultation with a specialist is recommended for interpretation and diagnosis*

Treatment

- There is no specific treatment for acute cases.
- Treatment of chronic cases should be done by a responsible physician in consultation with a hepatologist or infectious disease specialist.[5]
    - Treatment is individualized for those with chronic infection meeting criteria, when immunosuppressive therapy is being considered or for pregnant women to prevent perinatal transmission.

Reservoir

Humans.[1]

Transmission

Transmission occurs through mucosal or percutaneous contact with infected blood or body fluids. The most potentially infectious body fluids include blood, serum, vaginal secretions, semen, pericardial, amniotic, cerebrospinal, pleural, synovial, and peritoneal fluids. However, HBsAg has also been detected in human milk, tears, and saliva.[2]

The principal routes of transmission are:

- percutaneous (e.g., injection drug use, exposure to blood or body fluid),
- sexual (e.g., vaginal, anal, or oral),
- vertical (i.e., birth parent to infant during pregnancy, or childbirth), and/or
- horizontal (e.g., between household or close contacts through skin lesions, or sharing of blood-contaminated toothbrushes and razors; this is a common route of infection between young children in endemic countries).

Infections also occur in settings of close personal contact through unrecognized exposure to infectious body fluids. HBV is stable on environmental surfaces for at least seven days and can therefore be transmitted indirectly via inanimate objects.[1]

In Canada, the risk of transmission from screened and donated blood, manufactured blood products, and transplanted organs is minimal due to donor screening and the processing of blood products.[4]

Incubation Period

The incubation period for acute infections is 45 to 180 days, with an average of 60 to 90 days. HBsAg may appear in as little as two weeks, but in rare cases, as long as six to nine months. The variation is related in part to the amount of virus in the inoculum, the mode of transmission, and host factors.[1]

Period of Communicability

Individuals who are HBsAg positive are considered infectious, and the risk of transmission is highest during the acute phase of illness. HBsAg can be detected one to two months before and after symptom onset. Chronic HBV cases remain infectious indefinitely.[1,4]

Host Susceptibility

Susceptibility is universal. Protective immunity follows infection if HBsAg disappears, and hepatitis B surface antibody (anti-HBs) develops. Individuals who do not have documented history of receiving a valid and age appropriate series of hepatitis B vaccine, or who do not have a protective level of anti-HBs (≥10 IU/L), are considered susceptible.[2,4] For more information on risk factors that increase susceptibility, refer to Table 3: Risks for Hepatitis B Infection.

Table 3: Risks for Hepatitis B Infection

Risk Factors	Lifestyle Risk
• Immigration from or to a known hepatitis B endemic country (prevalence ≥ 8%) • being born to a person who is HBV positive • Household and sexual contacts of a known hepatitis B acute or chronic case • Household or close contacts of children adopted from a known hepatitis B endemic country (prevalence ≥ 8%) • Current or past history of incarceration (includes staff who work in correctional facilities) • Frequent receipt of blood or blood products (e.g., hemophiliacs) • Emergency service workers, HCWs, and others with potential occupational blood or body fluid exposure • History of chronic renal disease including receipt of chronic dialysis • HIV infection • Individuals with congenital immunodeficiencies	• Engaging in high risk sexual activities e.g.: - having unprotected sexual contact with someone who is HBV positive - unprotected sexual contact with new partners - having multiple sexual partners in the previous six months • Current or past history of sexually transmitted infection (STI) • Men who have sex with men (MSM) • Injection drug use (current or past history) and sharing of needles and other drug use paraphernalia (e.g., cookers, straws, pipes)

Incidence

Acute hepatitis B cases became reportable in Alberta in 1969 and chronic cases in 2008. Immigrants from hepatitis B endemic countries (prevalence ≥ 8%) account for most of the chronic cases reported in Alberta. For more information on current incidence rates and case counts of hepatitis B in Alberta, refer to the **Interactive Health Data Application**.

Public Health Management

Key Investigation

- Confirm that the individual meets case definition (acute or chronic infection).
- Assess risk factors and lifestyle factors for hepatitis B infection.
- Assess for co-infection with other blood-borne infections.
- Determine hepatitis B immunization history.
- Determine pregnancy status.
- Determine occupation where there is potential for occupational blood/body fluid exposure events (e.g., HCW[(C)])
- Identify household and other close contacts (as defined in Table 4: Definition of Close Contact below) with potential blood/body fluid exposure:

Table 4: Definition of Close Contact

Close Contacts
- persons living in the household, - needle sharing partners, - persons who share personal care items (e.g., razors, toothbrushes), - short- and long-term sexual partners, and - persons with other blood/body fluid exposure (e.g., unprotected first aid)

Management of Acute and Chronic Case

- Hospitalized acute and chronic cases should be managed in accordance with hospital infection prevention and control (IPC) guidelines.
- Provide education on signs and symptoms of disease, modes of transmission, and ways to reduce infection risk to others.
- Health care worker (HCW)[(C)] with acute/chronic hepatitis B should be assessed by the Zone MOH for potential risks of transmission to patients/clients. Those who perform, or may perform, exposure-prone procedures should be referred to the Alberta Expert Review Panel for Blood Borne Viral Infections in Health Care Workers for further assessment and recommendations.
- Recommend that pregnant individuals who are a confirmed HBV acute/chronic case should consult their physician for referral to a hepatologist or infectious disease specialist for assessment and follow-up during pregnancy to decrease possible transmission to the infant.
- Recommend that all chronic cases should consult their physician for referral to a hepatologist or infectious disease specialist for assessment and management recommendations.
- Discuss the importance of identifying any contacts that may have been exposed so that public health follow-up of those contacts can be arranged.
- Provide information on community support agencies and how to access support and services.
- Refer to Table 5: Additional Management Recommendations

[(C)] HCWs are defined as: all health practitioners and all individuals (including nutrition and food services, housekeeping, recreation etc.) at increased risk for exposure to, and/or transmission of, a communicable disease because they work, study, or volunteer in one or more of the following health care environments: hospital, nursing home (facility living), supportive living accommodations, or home care setting, mental health facility, community setting (e.g. paramedics, EMS, firefighters, police officers), office or clinic of a regulated health professional (i.e., this includes professionals regulated under the *Health Professions Act* and the *Veterinary Profession Act*) clinical laboratory.

Table 5: Additional Management Recommendations

Acute Cases	Chronic Cases
• Recommend testing for HBsAg and anti-HBs after six months to determine if case has cleared the infection or whether a chronic case • status has developed. • If testing has been completed sooner and loss of HBsAg and positive anti-HBs is demonstrated, repeat testing at six months is not necessary. • If serology results at six months show HBsAg and Anti-HBs are negative, the case may potentially be in the window period.[D] • Recommend retesting at six-month intervals to determine whether infection has resolved (Anti-HBs develops) or has become chronic (HBsAg reappears).	• Recommend Hepatitis A vaccine and Pneumo-P (Pneumococcal Polysaccharide 23). Refer to the *current* *Alberta Immunization Policy (AIP)* for eligibility criteria • Further testing may be required to determine the extent of liver involvement and treatment options; medical management should be done in consultation with a specialist.

Management of Contacts

- Assess each close contact (refer to Table 4: Definition of Close Contacts) as follows:
  - Determine risk of past infection.
  - Serology may be required to determine status.
  - Recommendations for serology, post exposure prophylaxis (PEP), and vaccine depend on age, history, and type of contact. See the following tables:
    - Table 6A: Immunization and Serology Recommendations for Management of Close Contacts at Low Risk of Previous Hepatitis B Infection. *(**Note**: Low risk refers to contacts who are NOT from a hepatitis B endemic country (prevalence ≥ 8%) and/or do not have other risk factors for acquiring disease)*
    - Table 6B: Immunization and Serology Recommendations for Management of Close Contacts at High Risk of Previous Hepatitis B Infection. *(**Note:** High risk refers to contacts who are from a hepatitis B endemic country (prevalence ≥ 8%) and/or have other risk factors for acquiring disease)*
    - Table 7: Post-Exposure Prophylaxis for Infants Under 12 Months of Age

Post-Exposure Prophylaxis (PEP) for Contacts

- Hepatitis B vaccine is the most important intervention for post-exposure prophylaxis (PEP) for those who are susceptible, and a complete immunization series provides 90% protection against hepatitis B infection.
- Hepatitis B immunoglobulin (HBIG) may provide additional protection through immediate short-term passive immunity.
- PEP should be offered to susceptible individuals in the following circumstances:
  - infant born to birth parent who is an acute or chronic case,
  - sexual or household contacts of an acute or chronic case, and
  - percutaneous or mucosal exposure to blood or body fluids potentially containing HB virus.
  - For management of blood/body fluid exposures (BBFE) in occupational and community settings refer to the current Alberta guidelines for post-exposure management and prophylaxis: HIV, Hepatitis B, Hepatitis C and sexually transmitted infections.

[D] The window period of HBV infection refers to the time between when HBsAg disappears followed by the appearance of anti-HBs. During this window period, both HBsAg and anti-HBs can be undetectable.[2]

Table 6A: Management of Close Contacts at Low Risk* of Previous Hepatitis B Infection (Immunization and Serology Recommendations)

Immunization Status	Serology Result	Recommendations			
Unimmunized Test for: • anti-HBs • HBsAg	• anti-HBs positive[**] • HBsAg negative	Offer hepatitis B vaccine series for long-term protection Post-immunization serology not required			
	• anti-HBs negative • HBsAg negative	Offer two or three dose vaccine series[$] Additionally, offer HBIG to: • sexual contact if within 14 days after last exposure • non-sexual contact with blood/body fluid exposures if within seven days after last exposure	Post-immunization serology (anti-HBs and HBsAg) at least one month after last dose of vaccine and at least six months after if HBIG is given		
	• anti-HBs negative • HBsAg positive	Determine acute/chronic case status and follow-up as per guideline			
Immunized: Documented valid two[$] or **three** dose series (age appropriate and spaced appropriately) Test for: • anti-HBs • HBsAg	• anti-HBs positive • HBsAg negative	Consider immune No further follow-up required.			
	• anti-HBs negative • HBsAg negative	Offer one dose of vaccine Additionally, offer HBIG to: • sexual contact if within 14 days after last exposure • non-sexual contact with blood/body fluid exposures if within seven days after last exposure	Test anti-HBs and HBsAg Ensure testing is at least one month after the last dose of vaccine and at least six months after if HBIG is given	• anti-HBs positive	Consider immune No further follow-up required
				• anti-HBs negative • HBsAg negative	Complete second series of vaccine Test one month after vaccine series
				• anti-HBs negative • HBsAg positive	Determine if acute or chronic case and follow-up as per guideline
	• anti-HBs negative • HBsAg positive	Determine if acute or chronic case and follow-up as per guideline			
Immunized: Documented valid **two** complete vaccine series Test for: • anti-HBs • HBsAg	• anti-HBs positive • HBsAg negative	Consider immune No further follow-up required			
	• anti-HBs negative • HBsAg negative	Non-responder No further vaccine indicated	HBIG x 2. Give the second dose of HBIG one month after 1st dose	Test HBsAg 6 months after HBIG	
	• anti-HBs negative • HBsAg positive	Determine if acute or chronic case and follow-up as per guideline			
Immunized with incomplete vaccine series: **One or two** doses of three **dose** series OR One dose of two dose series[$] Test for: • anti-HBs • HBsAg	• anti-HBs positive • HBsAg negative	Complete hepatitis B vaccine series for long-term protection			
	• anti-HBs negative • HBsAg negative	Complete vaccine series Additionally, offer HBIG to: • sexual contact if within 14 days after last exposure • non-sexual contact with blood/body fluid exposures if within seven days after last exposure	Test anti-HBs and HBsAg. Ensure testing is at least one month after the last dose of vaccine and at least six months after if HBIG is given	• anti-HBs positive • HBsAg negative	Consider immune
				• anti-HBs negative • HBsAg negative	Second series of vaccine Test one month after vaccine series
				• anti-HBs negative • HBsAg positive	Determine if acute or chronic case status and follow-up as per guideline
	• anti-HBs negative • HBsAg positive	Determine if acute or chronic case and follow-up as per guideline			

* Low risk refers to contacts who are **NOT** from a hepatitis B endemic country (prevalence ≥ 8%) and/or do **NOT** have other risk factors as outlined in Table 3: Risk factors and Lifestyle Risks for Hepatitis B Infection

** Anti-HBs positive is ≥10 IU/L; negative is < 10 IU/L

[$] Two or three dose series as outlined in the current *Alberta Immunization Policy (AIP)*

Table 6B: Management of Close Contacts at High Risk* of Previous Hepatitis B Infection (Immunization and Serology Recommendations)

<table>
<tr><th>Immunization Status</th><th>Serology Result</th><th colspan="4">Recommendations</th></tr>
<tr><td rowspan="6">Unimmunized

Test for:
• anti-HBs
• HBsAg
• Anti-HBc</td><td>• anti-HBs positive**
• HBsAg negative
• anti-HBc negative</td><td colspan="4">Consider immune for this exposure
Offer hepatitis B vaccine series for long-term protection
Post-immunization serology not required</td></tr>
<tr><td>• anti-HBs negative
• HBsAg negative
• anti-HBc negative</td><td colspan="2">Offer two or three dose vaccine series$
Additionally, offer HBIG to:
• sexual contact if within 14 days after last exposure
• non-sexual contact with blood/body fluid exposures if within seven days after last exposure</td><td colspan="2">Post-immunization serology (anti-HBs and HBsAg) at least one month after last dose of vaccine and at least six months after if HBIG is given</td></tr>
<tr><td>• anti-HBs positive
• HBsAg negative
• anti-HBc positive</td><td colspan="4">Immune from past infection
No vaccine indicated</td></tr>
<tr><td>• anti-HBs negative
• HBsAg positive
• anti-HBc negative</td><td colspan="4">Determine if acute or chronic case and follow-up as per guideline</td></tr>
<tr><td rowspan="2">• anti-HBs negative
• HBsAg negative
• anti-HBc positive</td><td rowspan="2">Refer for HBV DNA testing</td><td>HBV DNA positive</td><td colspan="2">Follow-up as chronic case</td></tr>
<tr><td>HBV DNA negative</td><td colspan="2">Offer hepatitis B vaccine series</td></tr>
<tr><td rowspan="8">Immunized: documented valid two or three dose series$
(age appropriate and spaced appropriately)

Test for:
• anti-HBs
• HBsAg
• Anti-HBc</td><td>• anti-HBs positive
• HBsAg negative
• anti-HBc negative</td><td colspan="4">Consider immune
No further follow-up required</td></tr>
<tr><td rowspan="3">• anti-HBs negative
• HBsAg negative
• anti-HBc negative</td><td rowspan="3">Offer one dose of vaccine
Additionally, offer HBIG to:
• sexual contact if within 14 days after last exposure
• non-sexual contact with blood/body fluid exposures if within seven days after last exposure</td><td rowspan="3">Test anti-HBs and HBsAg.
Ensure testing is at least one month after the last dose of vaccine and at least six months after if HBIG is given</td><td>• anti-HBs positive</td><td>Consider immune
No further follow-up required</td></tr>
<tr><td>• anti-HBs negative
• HBsAg negative</td><td>Complete second series of vaccine. Test one month after vaccine series</td></tr>
<tr><td>• anti-HBs negative
• HBsAg positive</td><td>Determine if acute or chronic case and follow-up as per guideline</td></tr>
<tr><td>• anti-HBs positive
• HBsAg negative
• anti-HBc positive</td><td colspan="4">Immune from past infection
No additional vaccine indicated</td></tr>
<tr><td>• anti-HBs negative
• HBsAg positive
• anti-HBc negative</td><td colspan="4">Determine if acute or chronic case and follow-up as per guideline</td></tr>
<tr><td rowspan="2">• anti-HBs negative
• HBsAg negative
• anti-HBc positive</td><td rowspan="2">Refer for HBV DNA testing</td><td>HBV DNA positive</td><td colspan="2">Follow-up as chronic case</td></tr>
<tr><td>HBV DNA negative</td><td colspan="2">Consider immune
No further follow-up required</td></tr>
</table>

$ Two or three dose series as outlined in the current *Alberta Immunization Policy (AIP)*

<table>
<tr><th>Immunization Status</th><th>Serology Result</th><th colspan="4">Recommendations</th></tr>
<tr><td rowspan="6">Immunized: Documented and valid two complete vaccine series

Test for:
• anti-HBs
• HBsAg
• Anti-HBc</td><td>• anti-HBs positive
• HBsAg negative
• anti-HBc negative</td><td colspan="4">Consider immune
No further follow-up required</td></tr>
<tr><td>• anti-HBs negative
• HBsAg negative
• anti-HBc negative</td><td>Non-responder
No further vaccine indicated</td><td>HBIG x 2. Give the 2nd dose of HBIG one month after 1st dose</td><td colspan="2">Test HBsAg 6 months after HBIG</td></tr>
<tr><td>• anti-HBs positive
• HBsAg negative
• anti-HBc positive</td><td colspan="4">Immune from past infection
No additional vaccine indicated</td></tr>
<tr><td>• anti-HBs negative
• HBsAg positive
• anti-HBc negative</td><td colspan="4">Determine if acute or chronic case and follow-up as per guideline.</td></tr>
<tr><td rowspan="2">• anti-HBs negative
• HBsAg negative
• anti-HBc positive</td><td rowspan="2">Refer for HBV DNA testing</td><td>HBV DNA positive</td><td colspan="2">Follow-up as chronic case</td></tr>
<tr><td>HBV DNA negative</td><td colspan="2">Consider immune
No further follow-up required</td></tr>
<tr><td rowspan="8">Immunized with incomplete vaccine series:

One or two doses of three dose series
OR
One dose in a two dose series$

Test for:
• anti-HBs
• HBsAg
• Anti-HBc</td><td>• anti-HBs positive**
• HBsAg negative
• anti-HBc negative</td><td colspan="4">Consider immune for this exposure.
Complete hepatitis B vaccine series for long-term protection.
Post-immunization serology not required.</td></tr>
<tr><td rowspan="3">• anti-HBs negative
• HBsAg negative
• anti-HBc negative</td><td rowspan="3">Complete vaccine series.
Additionally, offer HBIG to:
• sexual contact if within 14 days after last exposure
• non-sexual contact with blood/body fluid exposures if within seven days after last exposure</td><td rowspan="3">Test anti-HBs and HBsAg
Ensure testing is at least one month after the last dose of vaccine and at least six months after if HBIG is given</td><td>• anti-HBs positive
• HBsAg negative</td><td>Consider immune</td></tr>
<tr><td>• anti-HBs negative
• HBsAg negative</td><td>Second series of vaccine
Test one month after vaccine series</td></tr>
<tr><td>• anti-HBs negative
• HBsAg positive</td><td>Determine if acute or chronic case and follow-up as per guideline</td></tr>
<tr><td>• anti-HBs positive
• HBsAg negative
• anti-HBc positive</td><td colspan="4">Immune from past infection
No additional vaccine indicated</td></tr>
<tr><td>• anti-HBs negative
• HBsAg positive
• anti-HBc negative</td><td colspan="4">Determine if acute or chronic case and follow-up as per guideline</td></tr>
<tr><td rowspan="2">• anti-HBs negative
• HBsAg negative
• anti-HBc positive</td><td rowspan="2">Refer for HBV DNA testing</td><td>HBV DNA positive</td><td colspan="2">Follow-up as chronic case</td></tr>
<tr><td>HBV DNA negative</td><td colspan="2">Consider immune
Complete hepatitis B vaccine series</td></tr>
</table>

* High risk refers to contacts who are from a hepatitis B endemic country with a high prevalence (≥ 8%) and/or have other risk factors as outlined in Table 3: Risk factors and Lifestyle Risks for Hepatitis B Infection

**Anti-HBs positive is ≥ 10 IU/L; negative is < 10 IU/L

$ Two or three dose series as outlined in the current *Alberta Immunization Policy (AIP)*

Table 7: Post-Exposure Prophylaxis for Infants Under 12 Months of Age

Infants born to HBsAg positive birth parent (acute or chronic cases) *	
Prophylaxis	**Indication**
Pre-immunization serology is not required	
HBIG	• Should be given as soon as possible, preferably within 12 hours of birth • Efficacy decreases significantly after 48 hours but may be given up to seven days after birth • The dose for newborns is 0.5 ml intramuscularly
Hepatitis B vaccine	• Dose is 0.5 ml intramuscularly and should be given at the same time as HBIG, but at different sites • Subsequent doses to complete series given as per the current *AIP* • Newborns born to a hepatitis B infected birth parent who weigh less than 2000 grams at birth should receive an additional dose of vaccine. See *AIP* for spacing.
Follow-up	• Infants born to a HBsAg positive birth parent should be screened for anti-HBs and HBsAg following completion of hepatitis B vaccine series. • Ideally testing should be done at least one month following vaccine and within six months of completion of series; however, post-exposure immunization serology testing for this group is recommended **after nine months of age** to avoid detection of passive anti-HBs from HBIG administered at birth and to maximize the likelihood of detecting late HB virus infection.

Infants less than 12 months of age whose birth parent or primary caregiver is an acute case	
Prophylaxis	**Indication**
Pre-immunization serology is not required	
HBIG	• HBIG should be offered as soon as possible and within seven days of last contact if the birth parent or primary care giver is an acute case or if the birth parent is a chronic case. • HBIG is NOT offered if the primary caregiver is a chronic case.
Hepatitis B vaccine	• Dose is 0.5 ml intramuscularly and should be given at the same time as HBIG, but at different sites • Subsequent doses to complete series given as per the current *AIP.*
Follow-up	• Infants should be screened for anti-HBs and HBsAg following completion of hepatitis B vaccine series. • Ideally testing should be done at least one month following vaccine and at least six months after HBIG.

Infants less than 12 months of age whose primary caregiver is a chronic case	
Prophylaxis	**Indication**
Pre-immunization serology is not required	
HBIG	• HBIG is not offered if the primary caregiver is a chronic case.
Hepatitis B vaccine	• Dose is 0.5 ml intramuscularly. • Subsequent doses to complete series given as per the current *AIP.*
Follow-up	• Infants should be screened for anti-HBs and HBsAg following completion of hepatitis B vaccine series. • Ideally testing should be done at least one to six months following vaccine.

* All pregnant individuals should be tested routinely for HBsAg and if testing was not done during pregnancy, it should be conducted urgently at the time of delivery.

HBIG should be given if it is highly suspected that the birth parent could be an acute or chronic case and/or has risk factors for HBV; consider hepatitis B vaccine if HBsAg results are not available within 12 hours of delivery.

Preventative Measures

- Hepatitis B containing vaccine is 95–100% effective in preventing chronic HBV infection for at least 30 years following immunization.
  - Susceptible individuals with high risk factors for infection, should be offered hepatitis B containing vaccine. Refer to the current *AIP* for vaccine eligibility and vaccine type.
  - Post immunization testing is recommended for certain individuals (e.g., infants born to a HBsAg positive person) if it is important to determine protection against potential ongoing exposure to HBV. Refer to the current *AIP* for recommendations.
- Prevent perinatal HBV by:
  - screening all pregnant individuals,
  - all pregnant individuals who are HBsAg positive should be referred to a hepatologist or infectious disease physician for assessment and possible intervention.
  - timely immunoprophylaxis should be provided to infants born to a HBsAg positive person. Refer to current *AIP* for more information.
- The following individuals should be routinely screened for HBV and offered hepatitis B-containing vaccine if found to be susceptible:
  - Individuals with multiple sexual partners or with a current/past history of STI,
  - Those infected with another blood-borne virus, and
  - Having current/past history of injection drug use.
  - Adopted children from countries or family situations in which there is a high prevalence of HBV
- Provide public education on hepatitis B prevention and risks.

Health Care Workers

- Under the *Health Professions Act (HPA)*, all health professionals are governed by regulatory colleges that are required to govern in a way that protects and serves the public interest. Accompanying standards of practices and codes of ethics guide the health care worker (HCW) responsibility for ensuring their competency to practice.
- HCWs with acute/chronic hepatitis B who are involved in exposure prone procedures (EPP) shall contact the Zone MOH or designate to discuss the potential risks of transmission to clients. Upon assessment by the Zone MOH, a worker may or may not be referred to the Alberta Expert Review Panel for Blood Borne Viral Infections in Health Care Workers (administered by the Alberta College of Physicians and Surgeons) for further assessment services, if indicated.
- The Panel's mandate is to review circumstances of HCWs found to have a blood borne viral infectious disease. The panel may receive referrals from MOHs regarding HCWs who perform exposure-prone procedures when there is uncertainty as to whether continued or modified professional practice is indicated.
- In general, the recommendations for management of a HCW with acute/chronic hepatitis B are outlined by the Public Health Agency of Canada's document, Guidelines on the Prevention of Transmission of Bloodborne viruses from Infected Healthcare Workers in Healthcare Settings.

References

1. Spradling P. Hepatitis, Viral. In: Heymann DL, editor. Control of Communicable Diseases Manual. 21st ed. Washington, DC: American Public Health Association; 2022.

2. American Academy of Pediatrics. Hepatitis B. In: Kimberlin DW, Barnett ED, Lynfield R, Sawyer MH, editors. Red Book: 2021 Report of the Committee on Infectious Diseases. 32nd ed. Itasca, IL: American Academy of Pediatrics; 2021.

3. Kodani M, Schillie SF. Hepatitis B - Vaccine Preventable Diseases Surveillance Manual | CDC [Internet]. 2020 [cited 2023 Feb 3]. Available from: https://www.cdc.gov/vaccines/pubs/surv-manual/chpt04-hepb.html

4. Government of Canada. Part 4-Active Vaccines- Hepatitis B vaccine: Canadian Immunization Guide [Internet]. 2017. Available from: https://www.canada.ca/en/public-health/services/publications/healthy-living/canadian-immunization-guide-part-4-active-vaccines/page-7-hepatitis-b-v

5. Hepatitis B - Chapter 4 - 2020 Yellow Book | Travelers' Health | CDC [Internet]. [cited 2023 Feb 9]. Available from: https://wwwnc.cdc.gov/travel/yellowbook/2020/travel-related-infectious-diseases/hepatitis-b

6. Garg R, Ramachandran K, Jayashree S, Agarwal R, Gupta E. Evaluation of blood samples collected by dried blood spots (DBS) method for hepatitis B virus DNA quantitation and its stability under real life conditions. J Clin Virol Plus [Internet]. 2022 [cited 2023 Jun 19];2:100111. Available from: https://doi.org/10.1016/j.jcvp.2022.100111

7. Villar LM, Bezerra CS, Cruz HM, Portilho MM, Flores GL. Applicability of Oral Fluid and Dried Blood Spot for Hepatitis B Virus Diagnosis. Can J Gastroenterol Hepatol [Internet]. Hindawi Limited; 2019 [cited 2023 Jun 19];2019. Available from: /pmc/articles/PMC6463598/

8. 2023-03-08 DBST case definition clarification Letter to MOHs.

9. Centers for Disease Control and Prevention (CDC). Interpretation of Hepatitis B Serologic Test Results [Internet]. 2023 [cited 2023 May 25]. Available from: https://www.cdc.gov/hepatitis/hbv/interpretationOfHepBSerologicResults.htm

10. World Health Organization (WHO). Training workshop on screening, diagnosis and treatment of hepatitis B and C.

11. Curry MP, Chopra S. Acute Viral Hepatitis. In: Mandell GL, Bennett JE, Dolin R, editors. Mandell, Douglas, and Bennett's Principles and Practice of Infectious Diseases. Philadelphia, PA: Churchill Livingstone Elsevier; 2010. p. 1577–92.

12. Chang JJ, Mohtashemi N, Bhattacharya D. Significance and Management of Isolated Hepatitis B Core Antibody (anti-HBc) in HIV and HCV: Strategies in the DAA Era.

Appendix 1: Hepatitis B Virus Infection – High Endemic Geographic Areas

Revision Date: January 26, 2018

Children younger than seven years of age whose families have immigrated to Canada from areas where there is a high prevalence (8% or higher) of hepatitis B are at increased risk of hepatitis B infections even if neither parent is a chronic carrier. These children are likely to be exposed to hepatitis B carriers through their extended families or when visiting friends and relatives in their country of origin and should be offered hepatitis B vaccine. Immunization can start with the routine vaccination schedule at two months of age, with the next doses to complete the series given at four and 12 months of age. Hepatitis B vaccine series can be started at any age (two months up to seven years of age) for children identified who meet these eligibility criteria.

Countries considered highly endemic (8% or higher HBsAg prevalence) for hepatitis B infection are listed by geographical areas below:

Africa (all countries except Algeria, Egypt, Libya, Morocco and Tunisia)

African countries with high hepatitis B endemicity

Angola
Benin
Botswana
Burkina Faso
Burundi
Cameroon
Cape Verde Islands
Central African Republic
Chad
Comoros
Congo
Côte d'Ivoire
Democratic Republic of the Congo
Djibouti
Equatorial Guinea
Eritrea
Ethiopia
Gabon
Gambia
Ghana
Guinea
Guinea-Bissau
Kenya
Lesotho
Liberia
Madagascar
Malawi
Mali
Mauritania
Mauritius
Mozambique
Namibia
Niger
Nigeria
Reunion Island
Rwanda
Saint Helena
Sao Tome and Principe
Senegal
Seychelles
Sierra Leone
Somalia
South Africa
South Sudan
Sudan
Swaziland
Togo
Uganda
United Republic of Tanzania
Western Sahara
Zambia
Zimbabwe

Central and Eastern Europe (including the independent states of the former Soviet Union) and the Middle East

Central and Eastern European and Middle Eastern countries with high hepatitis B endemicity

Albania
Armenia
Azerbaijan
Bulgaria
Denmark – Greenland (indigenous populations)
Georgia
Jordan
Kazakhstan
Kyrgyzstan
Malta
Republic of Moldova
Saudi Arabia
Tajikistan
Turkmenistan
Uzbekistan

Central and South America (interior Amazon basin and parts of the Caribbean)

Central and South American countries with high hepatitis B endemicity

Bolivia (Amazon Basin)	Dominican Republic	Peru
Brazil (Amazon Basin)	Haiti	Venezuela (Amazon Basin)
Columbia (Amazon Basin)		

North America

High hepatitis B endemicity occurs in the Alaska Native populations and indigenous populations in Northern Canada.

Southeast Asia and the South and Western Pacific Islands

Southeast Asian and Western Pacific countries with high hepatitis B endemicity

America Samoa Cambodia	Lao People's Democratic Republic	Taiwan Thailand
China (includes Hong Kong, Macao and Taiwan)	Marshall Islands Macao	Timor-Leste Tokelau
Cook Islands	Mongolia	Tonga
Easter Island	Myanmar (Burma)	Trust Territories of Pacific Islands
Federated States of Micronesia	Nauru	Tuvalu
Fiji	New Caledonia and Dependencies	Vanuatu
French Polynesia	Niue	Vietnam
Guam	Palau	Wallis and Futuna Islands
Hong Kong	Papua New Guinea	
Indonesia	Philippines	
Kiribati	Samoa	
Korea (North and South)	Solomon Islands	

References

[1] National Advisory Committee on Immunization. (2017). Canadian Immunization Guide (Evergreen ed). Ottawa, ON: Public Health Agency of Canada. www.canada.ca/en/public-health/services/canadian-immunization-guide.html

[2] World Health Organization (2012). *Hepatitis B, Disease distribution map.* Retrieved August 25, 2017 from: http://gamapserver.who.int/mapLibrary/Files/Maps/Global_HepB_ITHRiskMap.png?ua=1.

[3] Schweitzer, A., Horn, J., Mikolajczyk, R., Krause, G., & Ott, J. (2015). Estimations of worldwide prevalence of chronic hepatitis B virus infection: a systematic review of data published between 1965 and 2013. *The Lancet 386(1546-55)*. Retrieved August 25, 2017 from: http://www.thelancet.com/pdfs/journals/lancet/PIIS0140-6736(15)61412-X.pdf

[4] Gish, R. G., & Gadano, A. C. (2006). Chronic hepatitis B: Current epidemiology in the Americas and implication for management. *Journal of Viral Hepatitis, 13(12)*.

[5] American Academy of Pediatrics. (2015). *Red book: 2015 Report of the Committee on Infectious Diseases* (30th ed.). Elk Grove Village, IL: Author.